# Nguyen Xuan Cuong

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## Summary

Senior AI & Software Engineer with 5+ years of hands-on experience in AI and software engineering, specializing in building production-grade Generative AI agents, Advanced RAG architectures, and real-time Computer Vision systems. Proven track record in bridging deep learning research with scalable full-stack and cloud deployments (AWS, GCP, Azure, Hugging Face, Vercel, Supabase, Nosana). Adept at technical leadership, architectural design, and rapid prototyping using modern AI-assisted engineering workflows. Fluent in English (IELTS 6.5 equivalent) for international technical collaboration and research.

## Education

**FPT University – Da Nang, Vietnam**
B.Sc. in Information Technology (AI Major) | GPA: 8.28/10.00 | *Recipient of 100% Full-tuition FPT University Scholarship.*

## Work Experience

**PiXerse Technology – Da Nang, Vietnam** Mar 2025 – Present
*Senior AI Developer / AI Architect*

- Architected 'Pixity', a real-time AI tourism agent ecosystem integrating transport, weather, and event APIs; won 2nd Runner-up at GDG DevFest MienTrung 2025.
- Designed Graph-RAG pipeline combining Neo4j graph databases and Pinecone vector search for spatial-temporal itinerary planning.
- Implemented multi-turn conversational agents with autonomous tool-calling capabilities using Gemini and OpenRouter APIs.
- Built scalable asynchronous backend microservices using FastAPI and Node.js; optimized containerized CI/CD workflows with Docker.

**Arcronics Solution – Da Nang, Vietnam** Jun 2024 – Feb 2025
*Senior AI Engineer / Embedded AI Specialist*

- Researched and trained deep learning models for malware classification and cyber attack-type detection.
- Quantized, pruned, and integrated lightweight AI models into hardware systems and microchips (Firmware, FPGA).
- Built automated MLOps pipelines for data preprocessing, feature extraction, model retraining, and continuous evaluation.

## Honors & Awards

**2nd Runner-up**: Hackathon GDG DevFest MienTrung 2025. | **Second Prize**: AI4SE 2025 Competition.
**4th Place**: Hackathon Nosana Builders Challenge: Agents 102. | **Superteam Honorable Mention**: Timelocked Wallet (2025).
**Finalist**: AI4LIFE 2024, Research Connection 2025, Research Festival 2025, and Grab The Beyond 2026.

## Research & Publications

**MAFIS**: Multi-Agent Imitation by learning and sampling from FactorIzed Soft Q-function.
**Identification of Plant Leaf Diseases**: Using CNN and SVM for precision agriculture.
**The GYM Exercise Classification Dataset**: 2D/3D pose analysis using CNN-LSTM and Mediapipe.
**Towards Sparse Adversarial Perturbations**: Gradient-Guided Approach for Efficient Image Attacks.

## Projects

**Manim Agent – Distributed Educational AI Storyboard & Video Platform** Oct 2023 – May 2024
*Role: Lead AI & Fullstack Architect / Team Size: 2 / Layers: FastAPI, Celery, Redis AOF, Docker Sandbox, React, Supabase*

- Architected an end-to-end multi-service AI system that transforms educational ideas into structured storyboards, generates executable Manim Python scripts, and renders HD mathematical videos.
- Implemented Human-in-the-Loop review flow via WebSockets and real-time state synchronization for scene-by-scene video previews.
- Isolated untrusted code execution using asynchronous Celery worker queues deployed within a protected Hugging Face Docker Space runtime.

**MathSolver – Multi-Agent Geometry Problem Solver & Motion Renderer** Feb 2023 – Sep 2023
*Role: Senior AI Engineer / Team Size: 3 / Layers: Multi-Agent AI System, OCR Engine, Manim Engine, FastAPI, Next.js*

- Developed a multi-agent system extracting geometric shapes and equations from images via OCR and generating step-by-step mathematical reasoning paths.

- Designed automated code generation pipeline translating proof steps into 2D/3D Manim motion graphics with multi-LLM fallback orchestration over OpenRouter APIs.
- Containerized backend services with Docker and Cairo/Pango system bindings to deliver real-time video compilation.

**Medagen – AI Patient Triage & Diagnostic Assistant** Jul 2022 – Jan 2023
*Role: Fullstack AI Developer / Team Size: 6 / Layers: LangChain ReAct Agents, Advanced RAG (Supabase pgvector/Qdrant), Computer Vision*

- Built a clinical diagnostic assistant utilizing LangChain ReAct Agent framework and Gemini for multi-turn symptom triage, differential diagnosis, and medical tool-calling.
- Architected an Advanced RAG system over medical knowledge bases using 768-dim vector embeddings (Supabase Vector / Qdrant), hybrid search, and semantic document chunking.
- Integrated Computer Vision models (Vision Transformers / custom CNNs) for dermatological lesion classification and multimodal symptom verification.

**MagicCV – Autonomous Resume Tailoring AI Agent** Jan 2022 – Jun 2022
*Role: AI Engineer / Team Size: 2 / Layers: Mastra Agent Framework, Supabase pgvector, LLM Ranking Engine, pdflatex Compiler*

- Created an autonomous CV tailoring agent (Nosana Builders Challenge finalist) that extracts candidate profiles from GitHub, LinkedIn, and YouTube APIs.
- Implemented 768-dim vector similarity search with 3-level fallback strategies and Gemini LLM ranking algorithms to match career data with job descriptions in 3 seconds.
- Rendered publication-quality PDFs via Nunjucks LaTeX templating; containerized with Docker and deployed to Nosana decentralized GPU compute network.

**Lab Management – Real-Time AI Computer Vision Dashboard** May 2021 – Dec 2021
*Role: Computer Vision & Software Engineer / Team Size: 5 / Layers: YOLOv8, ByteTrack, Janus WebRTC, WebSockets, Next.js*

- Deployed YOLO object detection and ByteTrack multi-object tracking models for real-time indoor occupancy tracking and crowd flow analytics.
- Streamed live video telemetry with Janus WebRTC servers and WebSockets, delivering low-latency spatial tracking analytics to Next.js dashboards.

## Skills

**Languages:** Python (5+ years / Expert), C++ (5 years), TypeScript / JavaScript (4+ years), SQL (5 years), Shell Script
**Language Proficiency:** English (IELTS 6.5 Equivalent – Professional working proficiency), Vietnamese (Native)
**AI/ML & Agent Frameworks:** PyTorch (5 years), TensorFlow, LangChain, LangGraph, Mastra, Model Context Protocol (MCP), ReAct Agents, Tool Calling, Prompt Engineering
**Generative AI & RAG:** Advanced RAG, Graph-RAG (Neo4j), Vector Databases (Qdrant, Pinecone, Supabase pgvector), 768-dim Embeddings, Ollama, vLLM
**Computer Vision & NLP:** Object Detection (YOLOv8/10), Pose Estimation (MediaPipe), Multi-Object Tracking (ByteTrack), CNNs, Vision Transformers (ViT), OCR
**Cloud Platforms & Deployment:** AWS, Azure, Google Cloud Platform (GCP), Hugging Face (Docker Spaces), Vercel, Supabase, Nosana Decentralized GPU Network
**Web, Database & DevOps:** Next.js, React, FastAPI, Node.js, WebSockets, Celery, Redis (AOF), Docker, Docker Compose, Janus WebRTC, Git (CI/CD)